



Nirbhay Singh

FinOps | Cloud Cost Governance
Savings Plans | Right-Sizing | AI Cost Mgmt

CONTACT

T	+48 886 202 296
@	nirbhaysingh1@live.com
in	linkedin.com/in/nirbhaysingh1
GH	github.com/nirbhays
M	medium.com/@nirbhaysingh1
W	cloudtoai.in
C	credly.com/users/nirbhaysingh
L	Warsaw, Poland

CORE SKILLS

FINOPS

FinOps Framework, Cloud Cost Allocation, Chargeback/Showback, Unit Economics, Cloud Billing APIs

COST OPTIMISATION

Spot/Preemptible, Reserved/Committed, Savings Plans, Right-Sizing, Auto-Scaling, Resource Lifecycle

CLOUD BILLING

AWS (Cost Explorer, CUR, Budgets), GCP (Billing, BQ export), Azure (Cost Mgmt)

COST TOOLING

TokenMeter, InfraCents, Infracost, Kubecost, CloudHealth

INFRASTRUCTURE

Terraform, Kubernetes, Docker, CI/CD, GitHub Actions, Jenkins

DATA & DASHBOARDS

BigQuery, Redshift, Grafana, Looker, Python, SQL

GOVERNANCE

Tagging, Multi-Account, Budget Alerts, Policy-as-Code, OPA

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Nirbhay Singh

FinOps & Cloud Cost Optimisation Architect | EUR 4M+ Documented Savings

EUR 4M+

Total Savings

\$45K

AWS Savings (Deloitte)

60%

AI Cost Reduction

50+

Accounts Governed

14+

Certifications

PROFESSIONAL SUMMARY

Cloud & FinOps Architect with 11+ years of cloud engineering and EUR 4M+ in documented cost savings at Bosch Europe. Treats FinOps as a governance discipline, not a one-time cost-cutting exercise -- unit economics, tagging standards, and, as AI workloads scale, token-level cost accountability. Built two open-source tools that operationalise that discipline: TokenMeter, which cut LLM spend 60% through provider-aware routing, and InfraCents, which prices Terraform changes on every pull request before they merge. Pairs deep multi-cloud architecture experience with the governance needed to keep the savings sustainable.

KEY ACHIEVEMENTS

- Delivered EUR 4M+ in documented cloud savings at Bosch Europe through FinOps governance, optimisation, and cost-aware architecture
- Cut AWS costs \$45,000 at Deloitte through Spot Instance fleet management, Reserved Instances, and workload right-sizing
- Open-sourced TokenMeter, a token-economics layer giving per-feature and per-team cost breakdowns, with smart routing cutting AI spend 60%
- Open-sourced InfraCents, pricing AWS+GCP cost impact on every Terraform pull request before it merges
- Built the enterprise FinOps practice behind that programme: chargeback/showback models, tagging standards, anomaly detection, and executive reporting

PROFESSIONAL EXPERIENCE



Cloud & AI Architect (FinOps Lead)

Dec 2022 - Present



Robert Bosch (Bosch Polska) Warsaw, Poland

- Built the FinOps programme behind EUR 4M+ in documented Bosch Europe savings: governance, optimisation, and cost-aware architecture
- Established chargeback/showback models, tagging strategies, and automated budget alerts across 50+ AWS and GCP accounts serving 15+ business units
- Built real-time cost dashboards and anomaly detection giving engineering and executive stakeholders shared visibility into spend
- Extended FinOps to AI workloads on Vertex AI and GKE: GPU scheduling, preemptible instances, and token-level cost tracking for LLM usage
- Trained 50+ engineers on cost-conscious architecture, embedding cost reviews directly into architecture governance



Senior DevOps Consultant

Jun 2021 - Nov 2022



McAfee

- Automated infrastructure with Terraform, cutting deployment overhead 50%+ and eliminating waste from manual provisioning
- Built cost-aware container orchestration across GKE, EKS, and AKS with autoscaling, pod resource limits, and Spot integration
- Built dashboards correlating infrastructure cost with performance metrics for data-driven optimisation
- Eliminated \$100K+ in unused resources through automated cleanup scripts and lifecycle policies



Senior DevOps Engineer

Dec 2018 - Jun 2021



Deloitte

- Cut AWS costs \$45,000 through Spot Instance fleet management, Reserved Instances, Savings Plans, and right-sizing

- Ran 1,000+ containers on AWS Batch with Spot Instances, cutting compute cost 70% for distributed batch processing
- Sped up CI/CD 30x, cutting build/test compute cost while improving developer productivity
- Built cost monitoring, anomaly detection, and budget alerting across AWS, GCP, Azure, and OCI

○ Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

■ PTC

- Designed cost-optimised AWS architectures: Auto Scaling, Spot Instances for non-critical workloads, Reserved Instances for baseline
- Managed 11,000+ server infrastructure, identifying and executing consolidation opportunities that cut operational cost
- Led cloud migration planning with TCO analysis, cost projection models, and ROI justification for executive stakeholders

○ System Engineer

Mar 2013 - Jul 2014

■ IBM

- Handled infrastructure operations and capacity planning for production environments

■ OPEN-SOURCE PROJECTS

TokenMeter -- Cost Intelligence Layer for LLM Apps

>> github.com/nirbhays/tokenmeter

One-line integration giving per-feature and per-team cost breakdowns across OpenAI, Anthropic, and Google -- tracks every token and routes by task complexity to cut AI spend up to 60%, at <5ms overhead.

STACK: Python, OpenAI API, Anthropic API, Google AI, Smart Routing, PyPI

InfraCents -- Terraform Cost Estimates on Every PR

>> github.com/nirbhays/infracents

Posts real-time AWS+GCP cost estimates on Terraform PRs before merge, so engineers see the exact cost impact of a change before it ships. Zero config.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Prevents costly security incidents by catching IaC misconfigurations pre-production across 100+ rules and 9 compliance frameworks, with AI-generated fix suggestions.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

■ CERTIFICATIONS & CREDENTIALS

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|---|---|
| ■ AWS Solutions Architect - Professional (Feb 2020) | ■ AWS Solutions Architect - Associate (Jan 2021) |
| ■ Google Professional Cloud Architect (Valid Jun 2026) | ■ Google Professional Cloud DevOps Engineer (Valid Dec 2027) |
| ■ Google Professional ML Engineer (Valid Dec 2027) | ■ Google Generative AI Leader (Valid Aug 2028) |
| ■ HashiCorp Terraform Associate | ■ CKAD - Certified Kubernetes Application Developer (Valid Sep 2026) |

■ TECHNICAL WRITING & THOUGHT LEADERSHIP

- "Our AI Bill Was \$4,800 Last Month -- Nobody Knew Why"
How engineering teams can reclaim control of LLM spend: per-feature cost breakdowns, smart model routing, and the story behind building TokenMeter.
<https://medium.com/@nirbhaysingh1>